

Ashwin Padaki

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SUMMARY

I use ideas and tools from theoretical computer science to understand and design algorithms for problems in machine learning and search. My recent work focuses on nearest neighbor search, quantization, and attention.

EDUCATION

University of Pennsylvania

Ph.D in Computer and Information Science

Philadelphia, PA

2024 – 2029 (*expected*)

Advisors: Erik Waingarten and Sanjeev Khanna.

Columbia University

B.A. in Computer Science, Mathematics

New York, NY

2020 – 2024

INDUSTRY EXPERIENCE

Research Scientist Intern

Pinecone

Jun 2026 – Aug 2026

New York, NY

- Built [VQ-bench](#), an open-source, composable vector quantization benchmark.

Quantitative Trader Intern

Optiver

Jun 2022 – Aug 2022

Chicago, IL

PAPERS

By convention, authors are listed in alphabetical order by last name.

1. **Ashwin Padaki**, Krish Singal, Erik Waingarten. [A Near-Optimal Space Lower Bound for Euclidean Diameter Estimation in Dynamic Streams](#).
(in submission)
2. Zachary Ives, Jiaming Liang, **Ashwin Padaki**, Erik Waingarten, Tian Zhang. [Prune, Don't Rebuild: Efficiently Tuning \$\alpha\$ -Reachable Graphs for Nearest Neighbor Search](#).
NeurIPS 2026
3. **Ashwin Padaki**, Amir Ingber, Edo Liberty. [VQ-bench: A Composable Vector Quantization Framework](#).
VecDB@VLDB 2026 (oral)
4. Sanjeev Khanna, **Ashwin Padaki**, Erik Waingarten. [Learning Partition Trees for Nearest Neighbor Search](#).
VecDB@VLDB 2026
5. Sanjeev Khanna, **Ashwin Padaki**, Erik Waingarten. [Sparse Navigable Graphs for Nearest Neighbor Search: Algorithms and Hardness](#).
SODA 2026

6. Sanjeev Khanna, **Ashwin Padaki**, Krish Singal, Erik Waingarten. [A Polynomial Space Lower Bound for Diameter Estimation in Dynamic Streams](#).
FOCS 2025

7. Henry Fleischmann, Kyrylo Karlov, Karthik C. S., **Ashwin Padaki**, Styopa Zharkov. [Inapproximability of Maximum Diameter Clustering for Few Clusters](#).
SODA 2025

8. Josh Alman, Yunfeng Guan, **Ashwin Padaki**. [Smaller Low-Depth Circuits for Kronecker Powers](#).
SODA 2023

TALKS

1. **VQ-bench: A Composable Vector Quantization Framework**
• VecDB@VLDB 2026 Workshop Sep 2026

2. **Learning Partition Trees for Nearest Neighbor Search**
• NYU Algorithms Seminar May 2026

3. **Sparse Navigable Graphs for Nearest Neighbor Search**
• SODA 2026 Conference Jan 2026
• NYC Graduate Student Theory Seminar Nov 2025
• DIMACS/Rutgers Theory Seminar Nov 2025
• CMU Theory Seminar Sep 2025

4. **Inapproximability of Maximum Diameter Clustering for Few Clusters**
• SODA 2025 Conference Jan 2025

SERVICE

Head Teaching Assistant <i>Algorithms for Big Data (UPenn)</i>	Sep 2025 – May 2026
Seminar Organizer <i>UPenn Theory Seminar</i>	Sep 2025 – present
Seminar Instructor <i>Columbia Undergraduate Learning Seminar in Theoretical Computer Science</i>	Sep 2023 – Dec 2023

ACHIEVEMENTS

National Science Foundation (NSF) Graduate Research Fellow	2024
Phi Beta Kappa Inductee	2024
Putnam Mathematical Competition, Top 500 Scorer	2022